Computer Fundamental

COMPLEMENT

Binary Number Divided into two part

i. Signed Binary Number ii. Unsigned Binary Number

i. Signed Number

- It is represent in +ve/-ve (+/-) integer.
- In sign bit, 1 bit reserve of sign bit (+ / -)

  0 → positive (+ve)

  1 → Negative (-ve)
- E.g. 0101 → +ve

  1101 → -ve

1's complement

- Range of n bit number is = $-(R^n-1)$ to $(R^{n-1}-1)$
- Here R= radix/base , n = bit
- E.g For 3 bit number = $-(2^3-1)$ to (2^3-1)

  = -3 to 3

2's complement

- Range of n bit number is = $-(R^{n-1})$ to $(R^{n-1}-1)$
- Here R= radix/base , n = bit
- E.g For 4 bit number = $-(2^{4-1})$ to $(2^{4-1}-1)$

  = -8 to 7

ii. Unsigned Number

- It is represent only +ve numbers.
- Each bit has 2 possible states = 0 or 1
- So for n bit, total combinations = 2^n
- Since counting starts form 0, the highest range of number is = 2^n-1.

✓ **Total values = 2^n**

✓ **Total range = 0 to 2^n-1**

Bits	Total values (2^n)	Range (0 to 2^n-1)
1 bit	2	0 to 1
2 bits	4	0 to 3
3 bits	8	0 to 7
4 bits	16	0 to 15
8 bit	256	0 to 255
32 bit	4294967296	0 to 4294967296
64 bit	18,446,744,073,709,551,615 (18.4 quintillion)	

Bit size	Type	Range
1bit	Unsigned	0 to 1
	Signed	0 to 0
2bit	Unsigned	0 to 3
	Signed	-2 to 1
4bit	Unsigned	0 to 15
	Signed	-8 to 7
8 bit	Unsigned	0 to 255
	Signed	-128 to 127
16 bit	Unsigned	0 to 65,535
	Signed	-32,768 to 32,767

- We can perform subtraction using the binary number system, so why were techniques like sign-magnitude, 1's complement, and 2's complement introduced.

- Because in a computer, there is **no separate circuit for subtraction**.
- A computer only has an **adder circuit**, and the adder performs all operations using addition — even(यहाँ तक की) subtraction.
- Therefore, a method was needed that could **convert subtraction into addition**, so that **one single adder circuit** could perform everything.
- Now, if In computers,

$$A - B = A + (\text{complement of } B)$$

This means: convert **B** into its negative form and then add it.

- The methods that perform this conversion are:
- My computer find 1's / 2's complement using Logic gates (AND, OR, NOT,NAND,NOR,XOR….).
- My computer solve Subtract ➔use two blocks (i)Adder circuit (Binary adder), (ii) Logic gates

In Computer, negative(-ve) numbers are represented using three methods.

(i) Sign - Magnitude **(ii) 1's complement** **(iii) 2's complement**

❖ Signed Magnitude

- Signed magnitude is a way to represent positive(+ve) and negative(-ve) number in binary by **using the leftmost** (most significant) bit as the sing bit. (leftmost bit देख कर पता करते हैं कौन -ve है कौन +ve)
- Here +ve sign use = 0 & -ve sign use = 1
- MSB = Most Significant Bit
- LSB = Least Significant Bit
- If Sign magnitude given that **n** bits then – 1st bit represent sign (-ve/+ve) and remainder (n-1) represent bits magnitude. Example

- ✓ 4 bits (0000) here 1st bit sing (0 then +ve) and remainder (4-1 = 3) bits represent bits magnitude.
- ✓ 6bits(100000) here 1st bit sing(1 then -ve) and remainder (6-1=5) bits represent bits magnitude.

- Example

To representations of zero +0 = 00000000 (8bit) -0 = 10000000	To representation of 25 +25 = 011001 (6bit) -25 = 111001
+18 = 00010010 (8bit) -18 = 10010010 MSB/leftmost LSB	+10 = 00001010, -10 = 10001010 Left Right

(+ve number को -ve बनाने के लिए MSB- Most significant bit में सिर्फ 1st वाले bit (इसे leftmost कहते हैं) को उल्ट देने से या change कर देने से वह -ve हो जाता हैं | 0 - positive , 1 - negative)

➢ Complements

- Complements are used in digital computers to simplify the subtraction operation and for logical manipulation.
- Complement refers to a mathematical operation or concept used in binary arithmetic,digital circuits and computer programming.

➢ Use of Complements

- In Subtraction
- Represent of negative number
- In Airthmetic logic unit (ALU)

➢ Real-life use of Complements

- For Subtraction in Digital circuits/calculators
- Store negative value in computer memory
- Check Error in CPU
- In networking

➢ Two types of Complement

1. 1's Complement 2. 2's Complement

1. 1's Complement

- Inverting/Changing all bits of a binary number (0 becomes 1, & 1 becomes 0).
- e.g. – Binary Number : 10110
  1's Complement : 01001

Q1. Find 1's Complement of decimal number 5 ?

→ Binary of 5 is = 101 ,

1's complement is = 010 Answer

Q2. Find 1's complement of decimal number +37 ?

→ Binary of (+37) = +100101 (+ के जगह 0 रख दें)

= 0100101

Now,find 1's complement this number 0100101 → 1011010

Q3. Represent -27 in 1's complement using 8 bits register.

→Binary (+27) = 11011 (in 8 bit → 00011011)

Now, 1's complement of 00011011 → 11100100

Q4. Find 1's complement of 110011

Convert 1's complement to Binary/decimal

Q5. 1010 (4-bit) Answer → -5

Check msb if 1 then -ve

Now -010 (010 → bit reverse)

Q6. 0011 (4bit) Answer → +3

Q7. 1100 (4bit) Answer → -3

Q8. 0000 (4bit) Answer → 0

Q9. 1110 (4bit) Answer → -1

2. 2's Complement

- Adding 1 (one) to the 1's Complement of a binary number.
- Number of bits is fixed

- e.g - Binary Number : 10110
  1's Complement : 01001
  01001 + 1 : 01010
  2's Complement : 01010 (solve)

Q. Convert Binary to 2's Complement

i) 10100 : 01100
ii) 101011 : 010101
iii) 110011 : 001101
iv) 100110 : 011010
v) 0011001100 : 1100110100

Difference sign magnitude, 1's complement, 2's complement

Let 5 in decimal and represent -5 using all three methods (4bits)

I. Sign-magnitude
   +5 = 0101 , sign bit =1, then -ve
   -5 = 1101
II. 1's complement
   +5 = 0101
   -5 = 1010
III. 2's complement
   +5 = 0101
   -5 =1010 + 1 = 1011

❖ Rule of 2's complement addition rule

Modern Computer use this technic for subtraction any number.

Steps

- Write numbers in binary (take one bit extra in MSB side)

- Convert negative number → **2's complement** (1's complement + 1)
- Add both
- If carry comes → **discard (MSB side)**
- Result MSB = 0 then +ve sign and MSB = 1 then find 2's complement again than final value is final answer with -ve sign

Q1. Add +9 and -5

Step1 → +9 = 01001 ---(i) & -5 =00101
Step2 → 2's complement of -ve number i.e. -5 = 11011-------(ii)
Step 3 → add (i) + (ii) ➔ 01001 + 11011 ➔ 00100 (carry को remove कर दो, i.e. discard)

Q2. (+7)+(-9) {5-bit} ➔ Answer (-2 →00010)
Q3. (-6) + (-5) {5 -bit} ➔ Answer (-11 → 01011)
Q4. (+12) +)-7) {6 bit} ➔ Answer (+5 → 000101)
Q5. (-8) + (+3) {5 bit} ➔ Answer (-5 → 00101)
Q6. (+15) + (-15) {6 bit} ➔ Answer (0)
Q7. (-10) + (+6) {5-bit} ➔ Answer (-4 → 00100)

❖ Rule of Multiplication

1. Decimal to Binary
2. Start with Multiplier (नीचे वाला) {bit 1 then write same number & bit 0 then write 0. Again every line left shift (*2) Example – bit 1 → number , bit 0 → 000
3. Add all rows
4. Result

Q1. 5 * 3

Step 1 ➔ 5 in binary = 101 , 3 in binary = 011
Step ➔ start with multiplier i.e. 011 (Read Right to left)
If First bit = 1 → 101 (same number)

If Second bit 1 (one left shift) → 101(same number)
If Third bit 0 (2 left shift) → 000
Step ➔ Add all 101
101 →(one left shift)
000 → (two left shift)
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1111
Step ➔ Answer 15 (1111)

========== HIRA KUMAR ==========